

Finding minimum of a function

```
using Optimization, OptimizationOptimisers, Zygote
```

```
f(x)=x[1]^2-2*x[1]+0.6;
```

```
x0=1.8*ones(1)
```

```
optf = OptimizationFunction((x,p)->f(x), ADTypes.AutoZygote())
```

```
prob = OptimizationProblem(optf, x0)
```

```
sol = solve(prob, Optimisers.Descent(0.1); maxiters=20)
```

```
retcode: Default
```

```
u: 1-element Vector{Float64}:
```

```
1.0115292150460684
```

Can change to

```
Optimisers.Momentum(0.1,0.9)
```

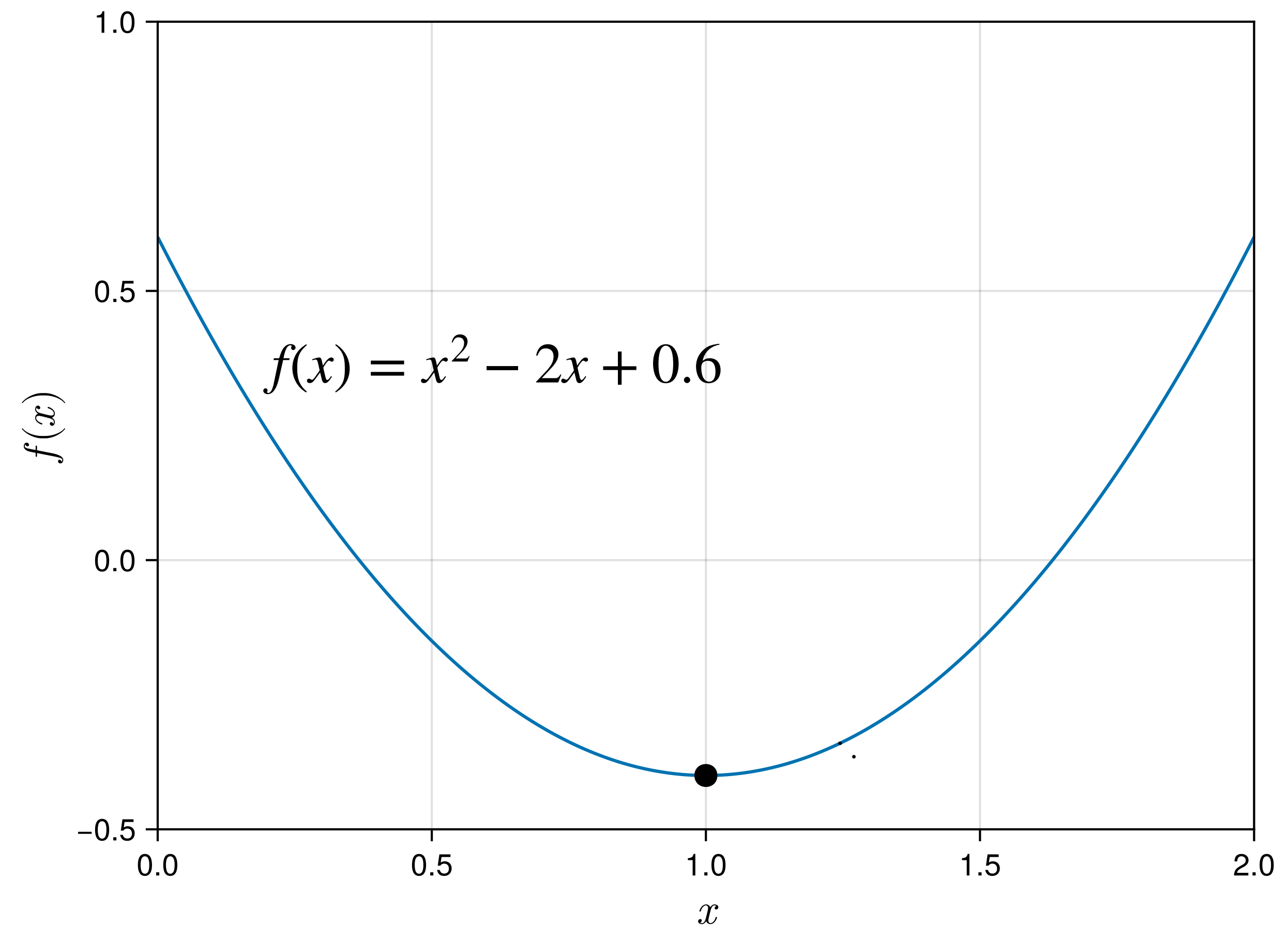
```
Optimisers.Nesterov(0.1,0.9)
```

```
Optimisers.Adam(0.1,(0.9,0.99))
```

Define function to optimize and its derivative (more later)

Define problem and tell the program the initial guess

Find the minimum



```

struct Point1D
    x::Float64
    fvalue::Float64
end
function my_callback1D(state, l)
    push!(vec_points, Point1D(state.u..., l))
    return false # continues the solver
end

vec_points=[] #Define empty vector to store intermediate solutions

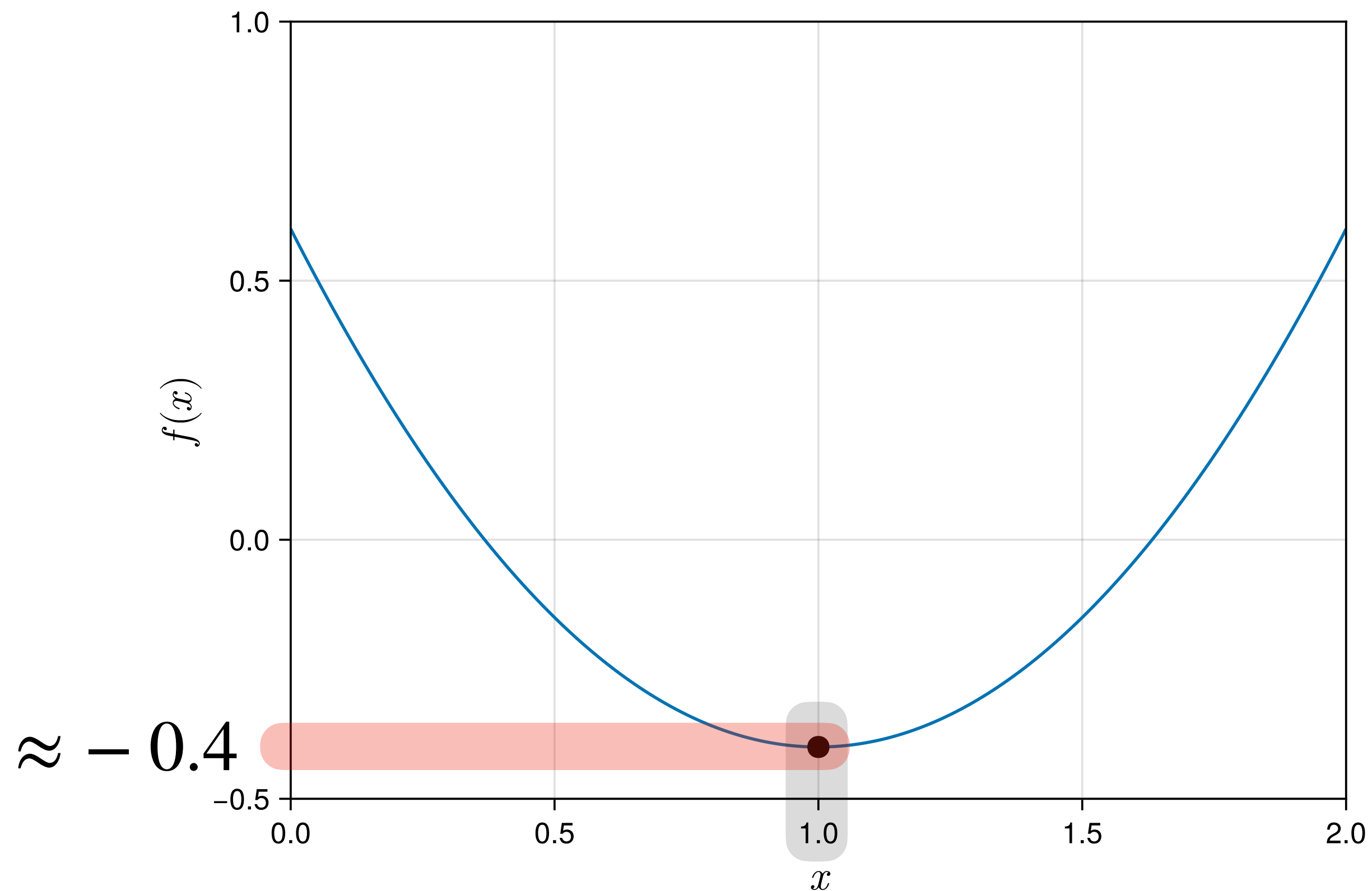
optf = OptimizationFunction((x,p)->f(x), ADTypes.AutoZygote())
prob = OptimizationProblem(optf, x0)
sol = solve(prob, Optimisers.Descent(0.1); maxiters=20, callback=my_callback1D);

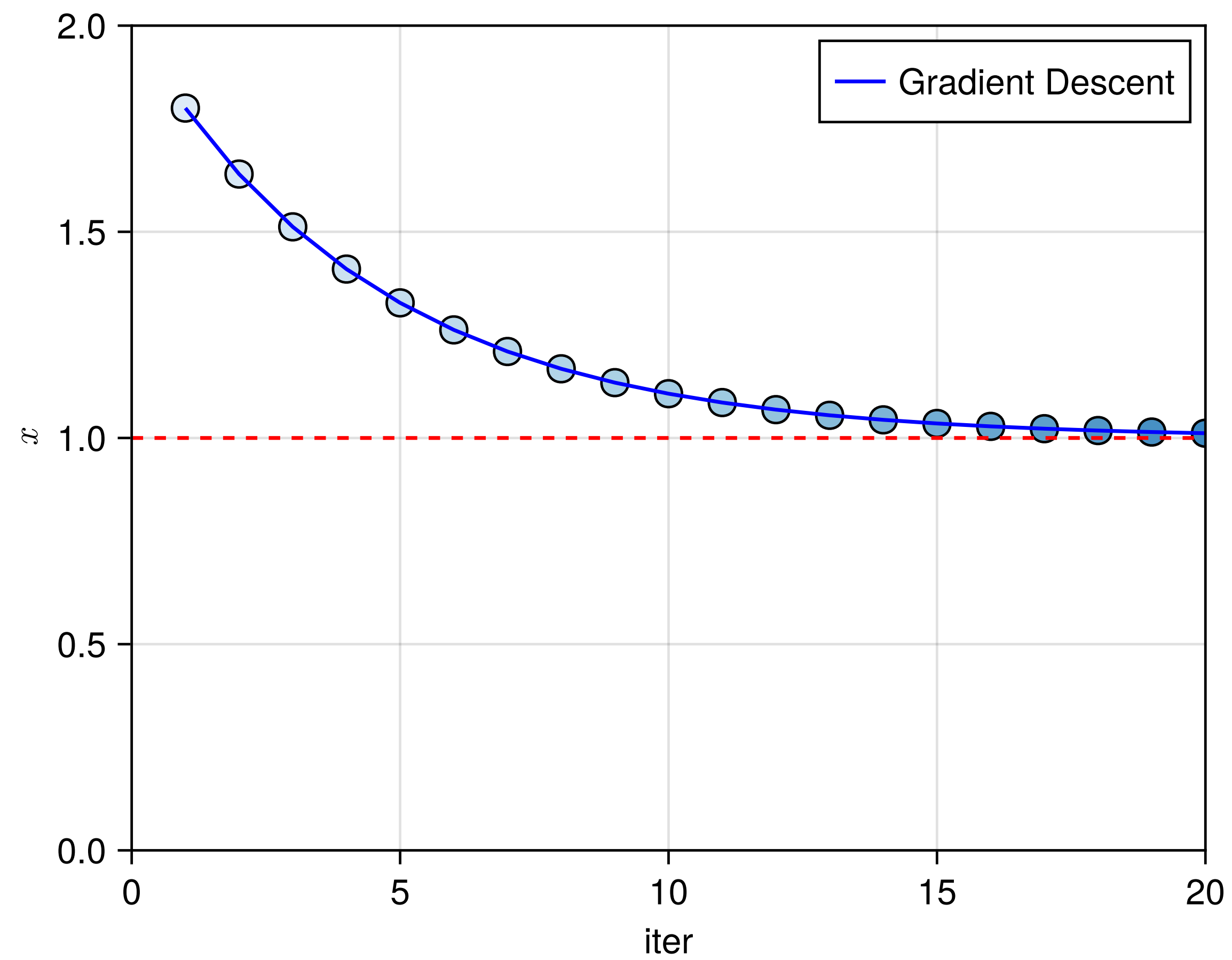
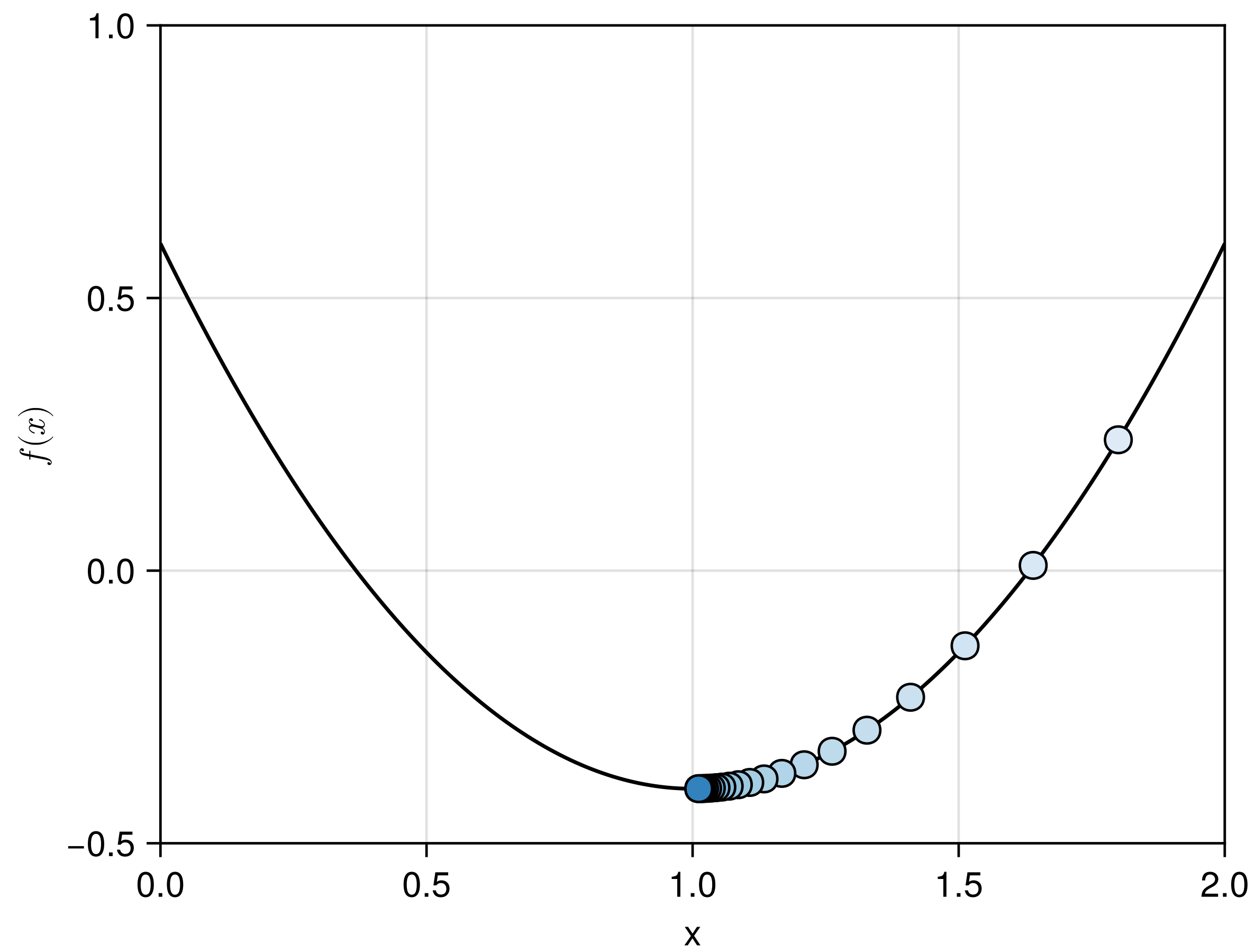
```

```

julia> vec_points
21-element Vector{Any}:
 Point1D{Float64}(1.8, 0.24000000000000001)
 Point1D{Float64}(1.6400000000000001, 0.0096000000000000164)
 Point1D{Float64}(1.512, -0.13785599999999987)
 Point1D{Float64}(1.4096, -0.23222784000000007)
 Point1D{Float64}(1.32768, -0.29262581759999995)
 Point1D{Float64}(1.262144, -0.33128052326400004)
 Point1D{Float64}(1.2097152, -0.35601953488896)
 Point1D{Float64}(1.16777216, -0.3718525023289344)
 Point1D{Float64}(1.1342177279999999, -0.38198560149051797)
 ⋮
 Point1D{Float64}(1.04398046511104, -0.3980657186886166)
 Point1D{Float64}(1.035184372088832, -0.39876205996071457)
 Point1D{Float64}(1.0281474976710656, -0.39920771837485736)
 Point1D{Float64}(1.0225179981368524, -0.3994929397599086)
 Point1D{Float64}(1.0180143985094818, -0.3996754814463416)
 Point1D{Float64}(1.0144115188075855, -0.3997923081256586)
 Point1D{Float64}(1.0115292150460684, -0.3998670772004215)
 Point1D{Float64}(1.0115292150460684, -0.3998670772004215)

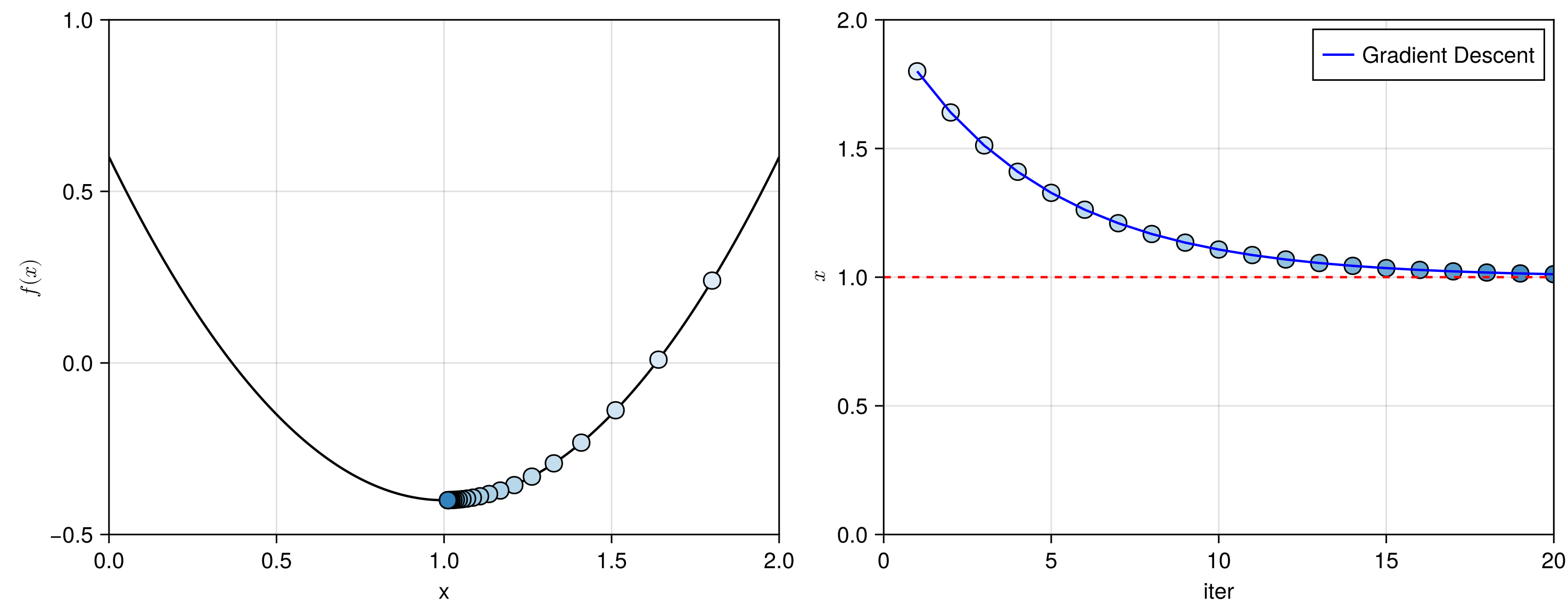
```





```
getproperty.(vec_points[:],:x)
```

```
21-element Vector{Float64}:  
 1.8  
 1.6400000000000001  
 1.512  
 1.4096  
 ⋮  
 1.0180143985094818  
 1.0144115188075855  
 1.0115292150460684  
 1.0115292150460684
```



```
getproperty.(vec_points[:],:fvalue)
```

```
21-element Vector{Float64}:  
 0.24000000000000001  
 0.0096000000000000164  
 -0.13785599999999987  
 -0.23222784000000007  
 ⋮  
 -0.3996754814463416  
 -0.3997923081256586  
 -0.3998670772004215  
 -0.3998670772004215
```